

CS104: Computer Networks and Protocols — Page 1: Network Architectures and Layering Models

1. OSI 7-Layer Reference Model vs. TCP/IP Architecture

Network protocol suites are organized hierarchically into distinct functional layers.

The ISO/OSI Reference Model defines 7 layers:

1. Physical Layer: Bit transmission over physical media (voltage levels, optical pulses).
2. Data Link Layer: Frame transmission, physical MAC addressing, framing, and CRC error detection.
3. Network Layer: Host-to-host packet routing, logical IP addressing, and path determination.
4. Transport Layer: End-to-end process-to-process communication, reliability, flow and congestion control.
5. Session Layer: Dialog management, session establishment, and checkpointing.
6. Presentation Layer: Syntax and semantics negotiation, data encryption, and compression.
7. Application Layer: Network interface for user applications (HTTP, DNS, SMTP).

The TCP/IP Internet model condenses this into 4 layers: Network Interface (Link), Internet, Transport, and Application.

CS104: Computer Networks and Protocols — Page 2: Network Layer, IP Addressing, and Routing

1. IPv4 Addressing and Subnetting

An IPv4 address is a 32-bit integer formatted as four dotted-decimal octets (e.g., 192.168.1.1). Classless Inter-Domain Routing (CIDR) uses variable-length prefix notation (/prefix) to allocate IP blocks efficiently.

Subnet Mask: A bitmask dividing an IP into Network ID and Host ID. For a /24 prefix, mask is 255.255.255.0, offering $2^{(32-24)} - 2 = 254$ usable host addresses (subtracting network address and broadcast address).

Address Resolution Protocol (ARP) dynamically resolves logical IP addresses to physical 48-bit Ethernet MAC addresses on a local subnet.

2. Routing Algorithms: Distance-Vector vs. Link-State

- Distance-Vector Routing (e.g., RIP): Routers iteratively share their entire routing table with immediate neighbors using the Bellman-Ford algorithm. Subject to the Count-to-Infinity problem, mitigated by Split Horizon and Poison Reverse.
- Link-State Routing (e.g., OSPF): Routers flood Link-State Advertisements (LSAs) across the entire network domain and independently compute shortest path trees using Dijkstra's algorithm. Avoids routing loops and converges rapidly.

CS104: Computer Networks and Protocols — Page 3: Transport Layer: TCP vs. UDP and Congestion Control

1. TCP vs. UDP Comparison

- UDP (User Datagram Protocol): Connectionless, lightweight, unacknowledged transport protocol with minimal 8-byte header overhead. Provides no guarantees for delivery, ordering, or duplicate protection. Ideal for latency-sensitive streaming, DNS queries, and real-time VoIP.
- TCP (Transmission Control Protocol): Connection-oriented, reliable byte-stream protocol. Implements connection setup via the Three-Way Handshake (SYN, SYN-ACK, ACK) and teardown via Four-Way Handshake (FIN, ACK, FIN, ACK).

2. TCP Reliability and Congestion Control

TCP ensures reliable transmission through sequence numbers, cumulative acknowledgments, and retransmission timers. Flow control is maintained through the Receive Window (rwnd) advertised in TCP headers to avoid buffer overflow on the receiver.

Congestion Control manages network bottleneck capacity via the Congestion Window (cwnd):

- Slow Start: cwnd starts at 1 MSS and doubles every RTT (exponential growth) until reaching ssthresh.
- Congestion Avoidance: cwnd increases linearly by 1 MSS per RTT (Additive Increase).
- Fast Retransmit and Fast Recovery: Upon receiving 3 duplicate ACKs, TCP infers packet loss, retransmits without waiting for timer expiry, and halves ssthresh (Multiplicative Decrease - AIMD).

CS104: Computer Networks and Protocols — Page 4: Application Layer Protocols: DNS, HTTP, and TLS

1. Domain Name System (DNS)

DNS is a distributed, hierarchical database mapping human-readable hostnames to IP addresses. The hierarchy comprises Root Servers, Top-Level Domain (TLD) servers (.com, .org), and Authoritative Name Servers.

Key DNS Record Types: A (IPv4 address), AAAA (IPv6 address), CNAME (canonical alias), MX (mail exchange server), and TXT (arbitrary text, SPF/DKIM verification).

2. HTTP Evolution and TLS Security

- HTTP/1.1: Introduced persistent TCP connections (Keep-Alive) and chunked transfer encoding, but suffers from Head-of-Line (HoL) blocking on single TCP streams.
- HTTP/2: Replaces textual format with binary framing. Multiplexes concurrent requests and responses over a single TCP connection, utilizes HPACK header compression, and supports server push.
- HTTP/3: Operates over QUIC (a UDP-based transport protocol), eliminating TCP Head-of-Line blocking across independent streams and supporting zero-RTT connection resumption.
- Transport Layer Security (TLS): Provides end-to-end encryption, server authentication via X.509 certificates, and message integrity via HMAC or AEAD ciphers.